

The Effect of Housing Supply Growth on Domestic Home Values

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ABSTRACT

This report attempts to analyze what causes the median sales price of new homes in the United States to change over time. The housing market is a product of the economic supply and demand model. However, home builders have struggled to meet demands, especially recently. As someone who grew up in Silicon Valley, I know how rapidly the market changes. Many people bought homes for relatively average prices in the 1980s and 1990s. Those homes can be worth millions in the present day. Major cities regularly experience housing shortages that can take years and different policies to stabilize. The 2008 global recession significantly impacted housing prices. Home values plummeted dramatically. The market eventually recovered. However, it is difficult for regression analysis methods to consider this. This report utilizes OLS and ARIMA model estimates to interpret how various economic factors changed the median home prices from 1968 to 2020. There will also be predictions made for periods after 2020.

INTRODUCTION

The major objective of this paper is to analyze how a change in housing supply affects housing prices in the United States. I also aim to use difference-in-difference analysis to compare the effects of policies passed in response to the 2008 housing crash. The financial crisis had widespread impacts on the housing market. Home values plummeted, and there was a soar in the home foreclosure rate because people started struggling to afford mortgage payments. This report is based on multiple data sources cited in the article, *Do Rent Controls and Other Tenancy*

Regulations Affect New Construction? Some Answers From Long-Run Historical Evidence. With this data, I intend to analyze the housing trends over time. The dependent variable I am interested in analyzing is the median sales price of new houses in the United States. Data for the number of new housing constructions is taken from another 2021 paper from Sebastian Kohl et al.

According to economic principles, a supply and demand model should determine housing prices. As the supply of housing increases, the demand should decrease, leading to a decrease in the median house price. Many cities have struggled to meet rising demand. For example, Silicon Valley and Santa Cruz have experienced a significant inflow of tech workers. Recent inflation levels exacerbate the frustration of high prices. The data set has multiple potential regressors, including real GDP, wages, and unemployment. This project will involve a null hypothesis test where the effect on new housing constructions, represented by β , is initially assumed to be zero. The explanatory variables are highly likely to be endogenous. This means the OLS method for calculating a β estimator will be biased and inconsistent. Economic policy not specified in the data can influence the housing supply. Some data manipulation is required for analytical purposes. The source data records data from 1870. However, I am only looking for data for consistency in 1968-2023. There are many countries in the source data. Still, since this is about analyzing the impacts on domestic housing markets, the data is converted into a subset consisting of only the records for the United States. Some unnecessary variables, such as the country letter codes, are removed in the subset. There are also some variables related to macroeconomic and microeconomic policies, like corporate debt, government bond rates, and loans to the private financial sector. I expect these variables to have a negligible effect on the dependent variable. They will likely be omitted in the regression equations used to determine the value of β . Another notable variable from the data is a dummy variable that indicates when the economy is in a

systemic financial crisis. For example, there may be a significant change in the data after the 2008 recession.

DATA

The data cited in the article consists of multiple countries beginning in the 19th century. I made an excel file that restricted the data to United States observations beginning in 1968. The only price variable in the original data is the nominal price set to 100 in the base year of 1990. For consistency and to avoid confusion from interpreting price, I replaced the nominal home prices with the Federal Reserve recorded median quarterly sales prices of new US homes. The yearly median price is inputted into the data after calculating the average of each financial quarter per year. We assume that the values are not adjusted for inflation. The median is more useful for collecting data and interpreting trends because it controls for higher cost-of-living areas skewing the data. The sample average housing price is \$149,530.2. A histogram of the housing prices shows a right-skewed distribution that peaks at approximately \$130,000. The major explanatory variable of interest is the number of new housing construction projects that start yearly. The years with the most construction should correspond to periods of high economic growth since high consumer confidence and favorable interest rates lead to more housing demand. The data consists of cross-sectional data, not panel data since different homes are examined at every new period. A plot of the average new construction per year shows a nonlinear trend. I was interested to see how the price changes over time. There was a decline between 2007 and 2008, the year of the housing market crash. Since 2010, housing construction has increased at a rate similar to 1990 - 2007, according to the plot in Figure 1. Figure 2 shows a plot of average housing construction and median price. At lower intervals, more constructions are shown to increase the nominal price. When the average new housing is greater than 100, the median price appears to be

split between intervals where it increases and decreases. Overall, there is a correlation coefficient of -0.43. As explained in Figure 3, median sales prices have increased yearly, except for the disruptions caused by the 2008 crash. The price is also highly correlated with years, with a coefficient of 0.98.

Prices didn't start increasing from the previous year until 2013. There are greater gaps between each data point after 2008. I intend to see whether this difference is statistically significant. The significant variables to be controlled include CPI, current account, money, and investment-to-GDP. Money refers to the money supply circulating in the economy. This makes sense as an important correlated variable because the Federal Reserve implements policies that change the money supply and stabilize the housing market. These variables are very likely to be endogenous, meaning that they are correlated with the other explanatory variables. For instance, unexplained factors that affect new home values in 1970 will also influence the values in 1971.

METHODOLOGY

The dataset consists of time-series data. The relationship between median price and new constructions is linear, which should make OLS valid for this data. OLS models fitted onto time series datasets assume no perfect collinearity and zero conditional mean. The OLS implications are evaluated with a 95% confidence interval. For this analysis, housing price is the dependent variable, and the average housing construction is the main regressor of interest. Housing prices are affected by many other factors besides supply and demand. The source paper does not discuss any major policy decisions meant to lower the prices. Before a policy is passed, the trend lines of treatment and control groups shift in parallel. However, the source data is only for the entire country and does not specify any housing differences between states. 2008 serves as an important cutoff point in the data. I intend to evaluate how price and construction compare

before and after the housing market crisis. Several policies were meant to address the challenges in the housing market. These include Federal Reserve monetary policies and new housing affordability programs

The coefficient represents the change in the index for a one-unit change in the independent variable. It's possible to manipulate the data by incorporating an inflation adjustment factor to convert the nominal index values into real values. The final equation is determined by the lasso regression method, which is very useful for this dataset because there are many variables, some of which may be redundant. It is also useful when there is multicollinearity in the data. The lasso regression kept the variables I expected, such as real GDP per capita, government expenditure, and the housing rental return. However, it also eliminates the average new supply, the main variable of interest. For this analysis, I keep the variables that can be used to indicate economic conditions. Considering the implications of a time series being weakly dependent or stationary is crucial. A series is weakly dependent if x_t and x_{t+h} are almost independent. The time series is not stationary because the sample mean does not stay constant over time. In other words, the mean varies over time. With an ADF test in R, we see a p-value of 0.035. Since we are interested in a 95% significance level, we reject the null hypothesis of non-stationarity. As a result, we assume the time series is stationary. OLS estimators are valid under this assumption. A weakly dependent process is also known as integrated of order zero, or I0. Unlike cross-sectional data, a time series raises major issues, such as time lags, the correlation over time, and serially correlated errors. ARIMA models are commonly preferred in a time series analysis. An ARIMA model combines autoregressive, differencing, and moving average methods together to reveal patterns in the data

RESULTS

I was first interested to know what β_1 is if a housing price regression is run on just the average housing construction. Interestingly, this produces an estimator of -1218, suggesting a negative relationship between prices and housing construction. According to this, as the number of new housing construction starts increases, we expect the median sales price to decrease by \$1218, holding all other variables constant. In reality, there are many correlated variables. When I run an OLS regression on average new housing along with the most relevant economic indicator variables, β_1 changes to 88.614. This is statistically significant. Many situations can explain a positive coefficient. For example, more construction can be a sign of economic growth. This leads to higher demand, which increases prices. The quality and location of new housing also play a significant role. Unfortunately, the original paper does not include information on the variation in housing quality. Expenditure, unemployment, wages, and the investment-to-GDP ratio are more significant than new construction. The significant variables with the highest magnitudes are investment-to-GDP, wages, and housing rental returns. The negative coefficient of wages is interesting because we expect higher wages to increase demand for higher-value homes. However, it is possible that the supply struggles to keep up with rising demand. The positive coefficient of unemployment seems counterintuitive. Although, high unemployment tends to correspond with the implementation of monetary policies that stimulate housing demand. The R^2 term of the OLS model is 0.998. A value close to 1 indicates that the observed data fits very closely to a linear model. A 95% confidence interval compares the estimates to the true sample mean price of 149530.2. A confidence interval plot of the OLS model shows that it struggles to fit a linear model. Even though some OLS estimators appear significant, the OLS method cannot help us draw any conclusions because there are too many factors for it to be consistent and unbiased.

The average rate of change in median price is 5644.9. The housing price increased by approximately units each year between 1968 and 2007. The average rate of change in 2012-2020 equals 11568.8 by comparison. It would be misleading to make any conclusions based on the rate of change because the periods are unequal, and the country was recovering from a recession at the time. We can use an ARIMA model to specify a predicted regression. This model is fitted to the pre-recession period. We can compare the predictions with real values to evaluate how well the regression model predicts the trajectory of housing prices. The median prices in 2021, 2022, and 2023 were \$396800, \$457475, and \$425150, respectively. I utilized an autoregressive integrated moving average model (ARIMA) to predict median prices over an additional 12 years after the dataset. Figure 7 shows the forecast model with the original time-series trend. The model predicts a steady increase in the median price. We calculate the forecast model's accuracy for the root mean square error. On average, the forecasts deviate from the observed values by \$6520.6. The forecast model predicts the prices in 2021-2023 to be \$348092.3, \$356589.7, and \$363828.4. A possible reason for this inaccuracy can be explained by the sudden increases in 2021 and 2022. Before 2021, the price each year had a small difference compared to the previous year. However, 2021 and 2022 have the largest differences over their previous years. For instance, the median price of a new home in 2021 increased by nearly \$60,000 from 2020. This period corresponds to the height of the COVID-19 pandemic, destabilizing the economy. The forecast model assumes that future years will follow the same patterns as before the pandemic. When considering OLS regression, I was interested in predicting how a forecasting model predicts housing prices. We create a training set from 80% of the data and a test set from the remaining 20%. A plot of the observed vs. predicted median home prices in Figure 8 suggests that the OLS model is inaccurate at predicting the response variable. In addition, the root mean

square error is 48280.58, a very high error term. If a null hypothesis assumes that the coefficient of new construction is zero, I expect a confidence interval plot of the OLS estimator to show a very high rejection rate. Assuming stationarity in the time series, OLS estimators are consistent but not necessarily unbiased. Although an ADF test indicates stationarity, home prices correlate with previous years. The dataset's variables can explain this correlation and other factors such as interest rates, demographic changes, and economic regulations. Another possible reason for the low accuracy is that the series is not stationary. An ARIMA model fitted to the data before 2008 has an AR1 coefficient of 1.37. This coefficient represents the effect of the previous observation on the current observation. The MA1 coefficient is -0.80. The negative sign means that past forecast errors negatively impact the current observations. The forecasted pre-recession model that ignores the effects of the 2008 recession predicts a steady linear increase. This does not accurately represent the real-world implications of economic conditions on the housing market. The plot in Figure 4 assumes that there will be no significant growth periods due to economic expansion or recession. A confidence interval of the ARIMA coefficient can be approximated with asymptotic theory, which uses standard errors to approximate the confidence intervals. The confidence interval plot in Figure 5 initially has no change in the coefficient estimate, but it increases when the coefficient index is 3. As mentioned previously, the pre-recession ARIMA model produces a coefficient of 1.37. The stable coefficient estimate can suggest that the parameters consistently affect the response variable. If this is the case, it aligns with home prices having a consistent linear increase until the recession. A sudden increase at index 3 could mean a sudden change in the effects of the explanatory variables. Unfortunately, it is challenging to determine the specific variables behind the change. A model diagnosis involves an analysis of the ARIMA model residuals. The residual plot shows regular intervals of positive and negative

residuals over time. The residuals tend to be larger in magnitude as time increases. The largest changes occur as the time series approaches 2008. After 2008, the positive residuals remained at the same level while the negative residuals became smaller in magnitude. This suggests that a forecasted model overestimates the sales price every other year. Also, the positive residual peaks stabilize close to zero. This indicates that the model is consistent with the observations based on most of the historical data. For 2021 and onwards, we expect the negative residuals to follow the same pattern since 2013, when they gradually get closer to zero. In recent years, the positive residuals have stayed at a relatively constant level of 10000

CONCLUSION

We cannot conclude that new construction starts significantly impact the sales value of a new home because there is a complex combination of other factors. OLS models do not accurately forecast pricing trends, even when controlled for as many explanatory variables as possible. The OLS estimators table suggests that variables like CPI(Consumer Price Index), expenditure, and the investment-to-GDP ratio are more important factors in determining a home's sales price at any given time. This would make sense because new construction is an endogenous variable determined by the multiple different variables in the data. The housing affordability programs introduced after 2008 may have helped the sharp decline in home values. Still, we cannot conclude on a difference-in-difference effect due to the high amount of correlated economic factors that can take time to have obvious impacts on construction or sales prices. Historically, public housing funding has been allocated to lower-income areas. If there was equal investment in middle and upper-class housing, a change in supply may have a more defined impact on making homes more affordable all around the country.

FIGURES

Figure 1

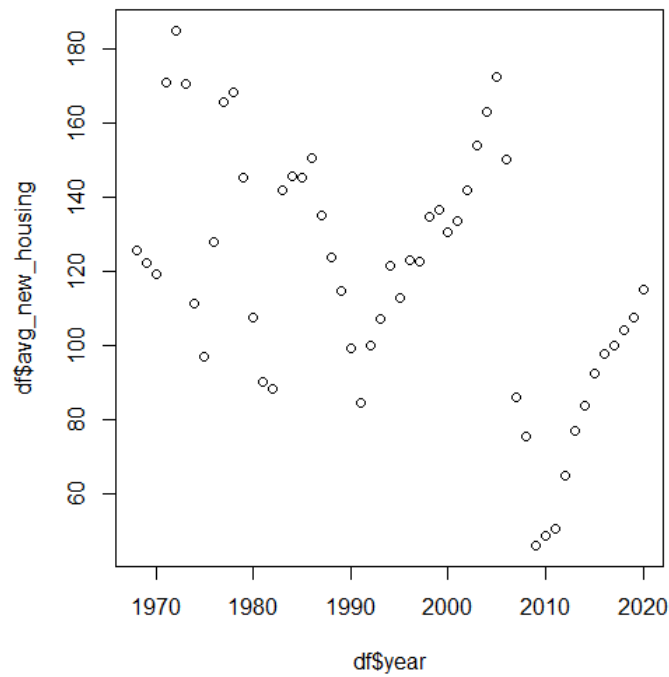


Figure 2

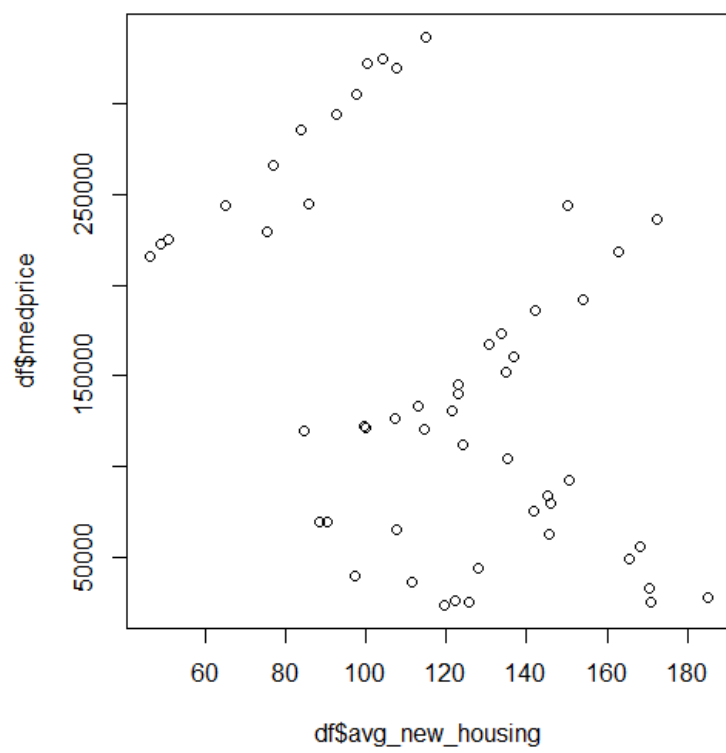


Figure 3

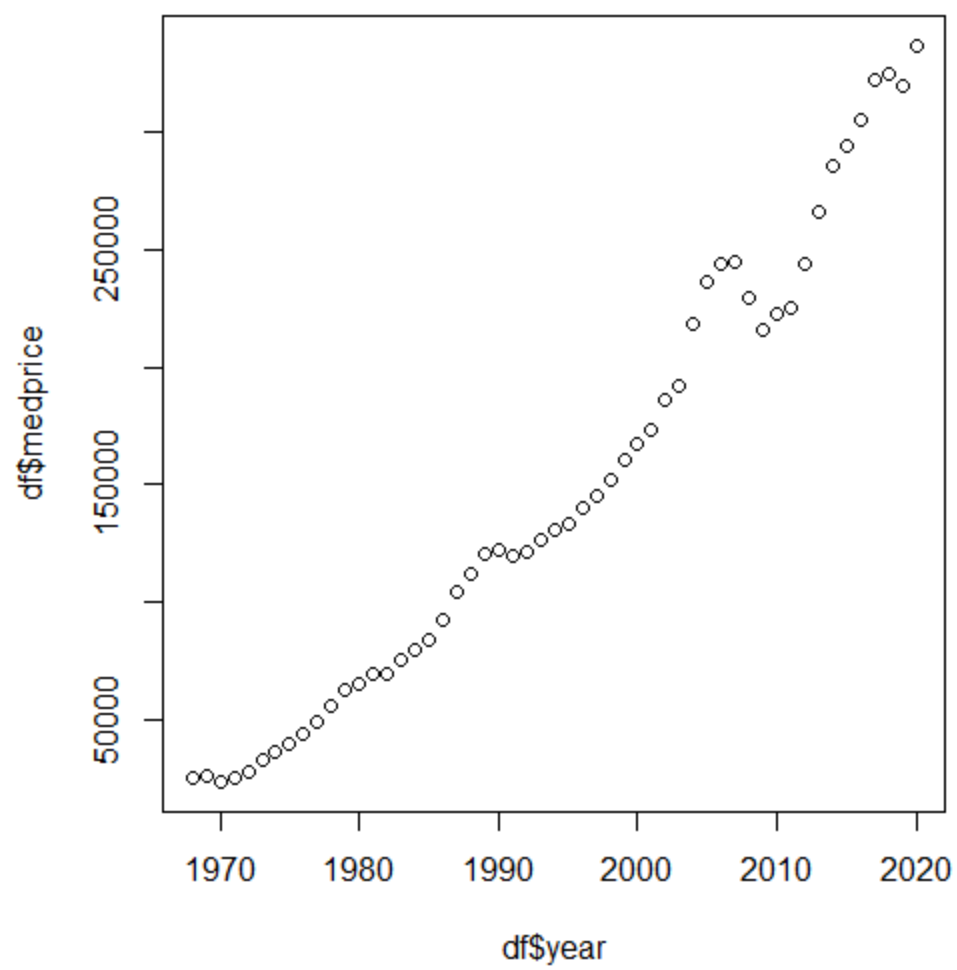


Figure 4

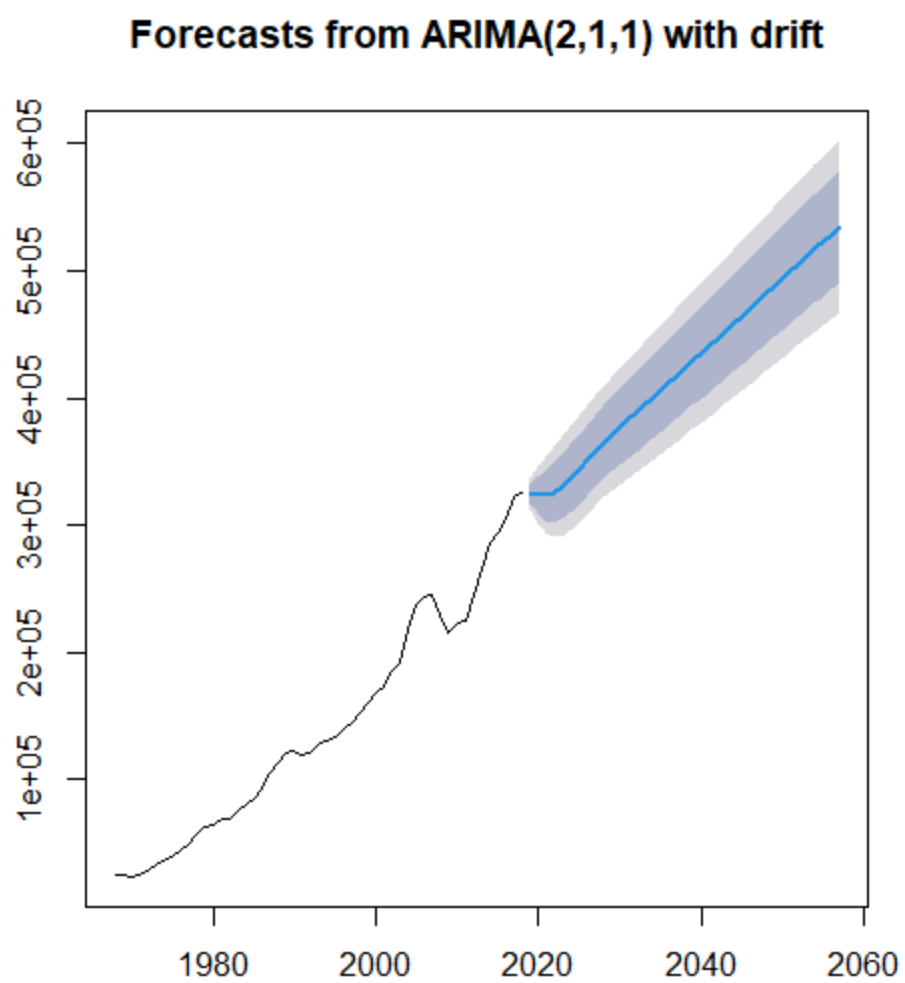


Figure 5

ARIMA Coefficient Estimates with 95% Confidence Inter

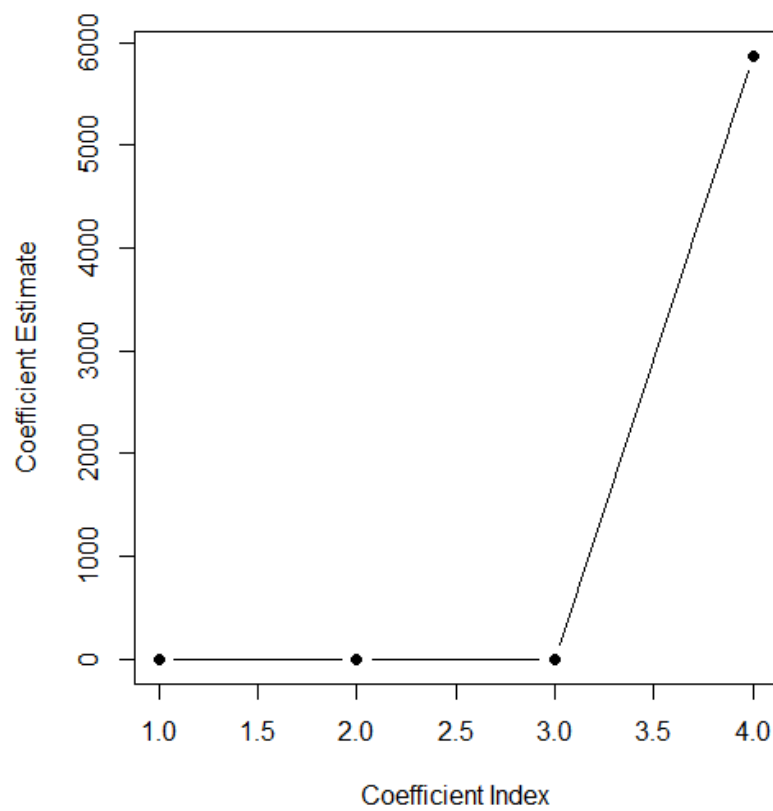


Figure 6

Residuals Plot of Pre-Recession ARIMA Model

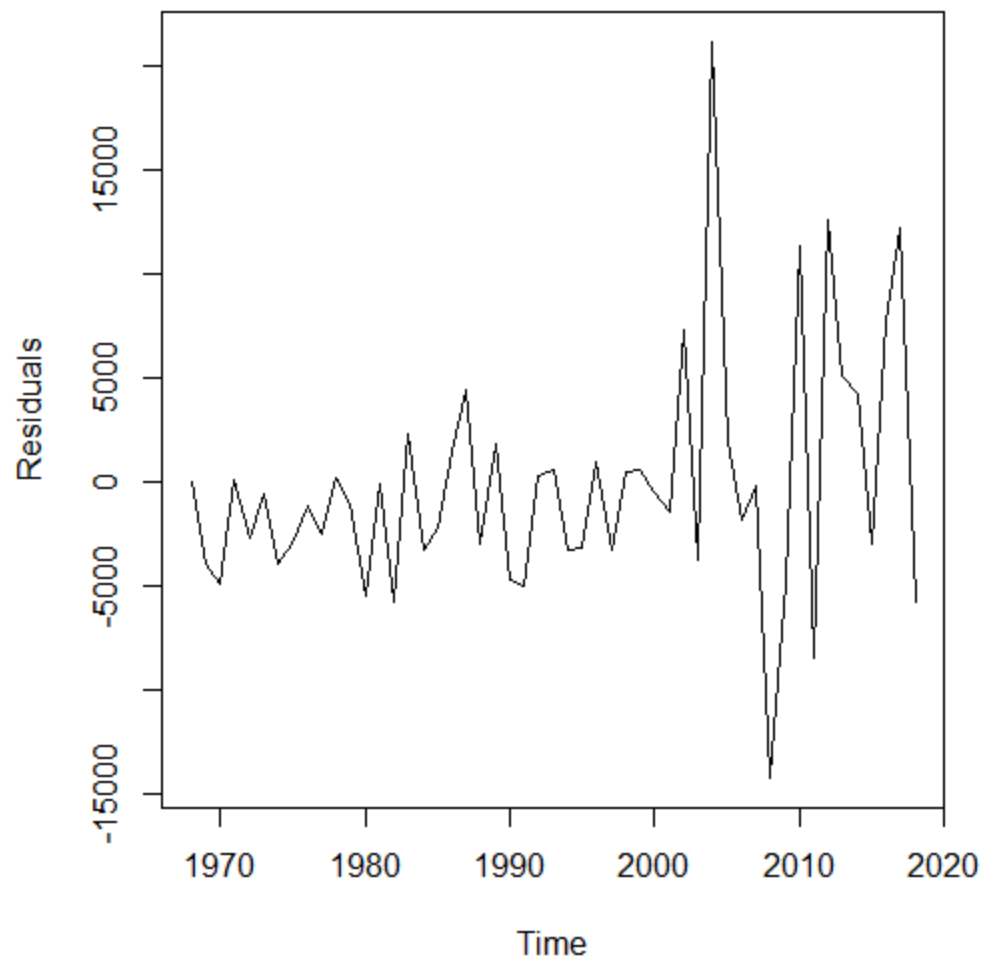


Figure 7

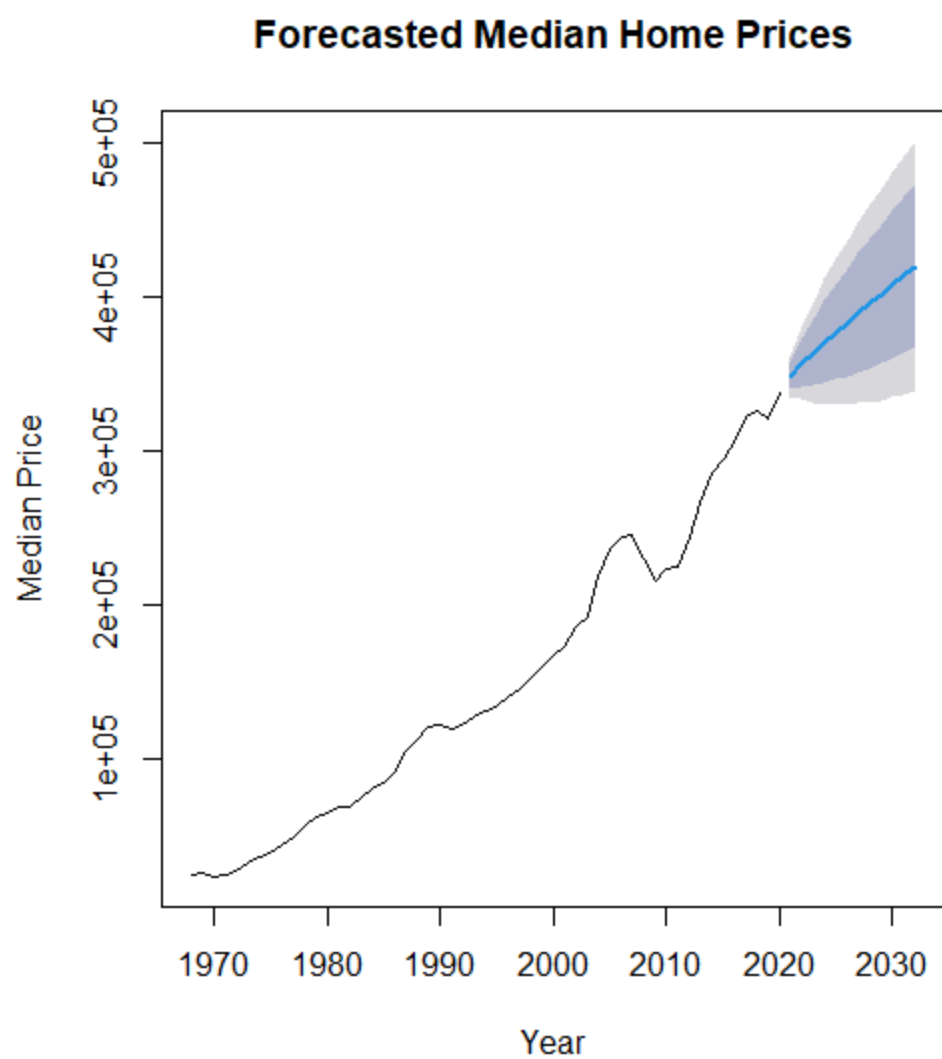


Figure 8

Observed vs Predicted

